

I used GitHub Copilot inside VS Code as an inline assistant. Copilot suggests code as I type, and I accept or reject each suggestion as I see fit. It helped me write this enhancement, but the final output you see is the outcome of my modifications. Because this is a maintenance task, I added the new channel in its own file, `whatsapp_service.py`, and left the engine (the `NotificationMedium` interface and the `AlertSystem` class in `notification_system.py`) unchanged. Copilot is driven by the surrounding code rather than a chat box, so the prompt was the new file plus the class declaration I wrote to guide it. In `whatsapp_service.py` I imported the existing interface and typed:

```
from notification_system import NotificationMedium class
WhatsAppService(NotificationMedium): def send(self, message: str) -> str:
```

Copilot suggested the body. I inspected and then accepted the return line after fixing it to the exact format the assignment requires: `[WhatsApp] Sending message: <message>`.

How I kept the signature matched to avoid rewriting

- I imported `NotificationMedium` from the engine module and subclassed it, so the new channel is bound to the one interface already in the system instead of a second copy.
- I rejected any Copilot suggestion that drifted from the interface, for example, versions that renamed a parameter or added extra arguments. The method had to stay `send(self, message: str) -> str` to match.
- Because `send` is marked abstract on the base class, Python will not instantiate a subclass that fails to implement it. I confirmed with `isinstance()` that `WhatsAppService` is recognized as a `NotificationMedium` before relying on it.
- I did not change the `AlertSystem` class or the `NotificationMedium` interface. The new channel is selected through the same `set_medium()` and `notify_user()` that were already there, and it is wired into the menu in the driver file (`notification_app.py`), not the engine. Adding a file rather than editing the engine keeps this a maintenance change rather than a rewrite.

Generating the test script I also used Copilot to write the test script, `maintenance_demo.py`, which proves the new channel works at runtime. I described the `AlertSystem` API in a comment that it takes a medium, has `set_medium()`, `notify_user()`, which returns a confirmation string, and `get_log()`, which returns a list, and let Copilot draft a `main()` that creates one `AlertSystem`, switches it across `EmailService`, `SMSService`, and `WhatsAppService` using `set_medium()`, and prints the shared log. I kept only the suggestions that went through those existing public

methods, so the test exercises the system the same way a real caller would, without reaching into AlertSystem's internals or modifying it.

Docstring adjustments

- I rewrote Copilot's comment into the same docstring style that I used for the rest of the project. A module header, a class docstring, and a method docstring with Parameters and Returns sections.
- I removed the suggestion's redundant inline comment that just repeated the docstring, and kept the return type annotation consistent with EmailService and SMSService.
- I noted in the module docstring that this is a Week 5 maintenance addition that imports and implements the existing interface.